

India's Agricultural Crop Production Analysis

TEAM ID	SWTID-2026-4021
PROJECT TITLE	India's Agricultural Crop Production Analysis
MAX MARKS	2 MARKS

CHAPTER – III

REQUIREMENT ANALYSIS

3.1 – Customer Journey Map

India's Agricultural Crop Production Analysis – Requirement Analysis & Customer Journey Map

Project Objective

To develop an Agricultural Crop Production Analysis system that helps farmers, agricultural departments, researchers, and policymakers monitor crop production, predict yields, identify risks, and make data-driven decisions for improving agricultural productivity and profitability.

Requirement Analysis

Functional Requirements

- Collect crop production data from various regions.
- Analyze crop yield trends and seasonal patterns.
- Monitor weather, soil, and irrigation conditions.
- Generate production forecasts and yield predictions.
- Provide crop-wise and region-wise reports.
- Track market prices and demand trends.
- Offer recommendations for improving productivity.
- Display insights through dashboards and visual reports.

Non-Functional Requirements

- User-friendly interface for farmers and stakeholders.
- Accurate and reliable data processing.

India's Agricultural Crop Production Analysis

- Secure storage of agricultural data.
- Fast report generation and analysis.
- Scalable system capable of handling large datasets.
- Mobile and web accessibility.

Customer Journey Map

Stage	Farmer Actions	Needs	Pain Points	System Solutions
Awareness	Learns about crop analysis tools through government programs, cooperatives, or agricultural officers	Information on crop planning and productivity	Lack of awareness and technical knowledge	Awareness and campaigns, training, and simple user guides
Planning	Selects crops based on season, soil, and market demand	Reliable recommendations and forecasts	Uncertain weather and market prices	AI-based crop recommendations and weather forecasts
Cultivation	Prepares land, plants seeds, and manages crops	Real-time farming guidance	Pest attacks, diseases, resource shortages	Alerts for pests, diseases, and irrigation management
Monitoring	Tracks crop growth and field conditions	Continuous crop health monitoring	Difficulty identifying crop problems early	Dashboard with crop health and production insights
Harvesting	Harvests crops and prepares produce for sale	Yield estimation and harvesting guidance	Post-harvest losses and labor issues	Yield prediction and harvest scheduling support

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Stage	Farmer Actions	Needs	Pain Points	System Solutions
Marketing	Sells crops in local markets or online platforms	Fair prices and market access	Price fluctuations and middlemen dependency	Market price analysis and digital marketplace integration
Evaluation	Reviews profits, yields, and performance	Performance reports and future planning	Limited data for decision-making	Detailed reports and improvement recommendations

Key Insights

- Farmers require timely information, accurate forecasts, and market intelligence.
- Weather uncertainty, rising input costs, and price fluctuations are major challenges.
- Data-driven agricultural analysis can improve productivity, reduce risks, and increase farmer income.
- Integration of technology, analytics, and government support can create a more sustainable agricultural ecosystem.

Expected Outcome

An efficient Agricultural Crop Production Analysis system that supports informed decision-making, enhances crop yields, improves market opportunities, and contributes to sustainable agricultural development in India.

BRAVO
— BUILDERS —